OS UNIT4 ASSIGNMENT

https://github.com/Srujanakenche/PES1UG24CS227-pes-vcs

PHASE 1:

Screenshot 1A — capture ./test_objects output (shows all PASS)

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % make all
gcc -Wall -Wextra -O2 -I/opt/homebrew/opt/openssl@3/include -c tree.c -o tree.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto -L/opt/homebrew/opt/openssl@3/lib
gcc -o test_objects test_objects.o object.o -lcrypto -L/opt/homebrew/opt/openssl@3/lib
gcc -o test_tree test_tree.o object.o tree.o index.o -lcrypto -L/opt/homebrew/opt/openssl@3/lib
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % ./pes init
Initialized empty PES repository in .pes/
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

Screenshot 1B — capture the sharded directory listing

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % find .pes/objects -type f

.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
```

PHASE 2:

Screenshot 2A — capture ./test_tree output (shows all PASS)

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % ./test_tree

Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

Screenshot 2B — capture the xxd hex dump of the tree object

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % ulimit -s unlimited
rm -rf .pes
./pes init
echo "hello" > file1.txt
./pes add file1.txt
./pes commit -m "test"
find .pes/objects -type f | while read f; do if head -c 4 "$f" | grep -q "^tree"; then echo "$f"; xxd "$f" | head -20; break; fi; done


Initialized empty PES repository in .pes/
Committed: d6b371f8ffe1... test
.pes/objects/09/b576c2eb51b78fcf9b8493a0909522a75f8ee86df0e926312deeb5fbe7dfb1
00000000: 7472 6565 2034 3900 3130 3036 3434 2066  tree 49.100644 f
00000010: 696c 6531 2e74 7874 002c f8d8 3d9e e295  ile1.txt.,..=...
00000020: 43b3 4a87 7274 21fd ecb7 e3f3 a183 d337  C.J.rt!........7
00000030: 6390 25de 576d b9eb b4                   c.%.Wm...
```

PHASE 3:

Screenshot 3A — capture the full sequence output with status

```
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % rm -rf .pes hello.txt file1.txt file2.txt bye.txt
  ./pes init
  echo "hello" > file1.txt
  echo "world" > file2.txt
  ./pes add file1.txt file2.txt
  ./pes status

  Initialized empty PES repository in .pes/
  Staged changes:
    staged:     file1.txt
    staged:     file2.txt

  Unstaged changes:
    (nothing to show)

  Untracked files:
    untracked:  index.h
    untracked:  report_assign4.pdf
    untracked:  object.c
    untracked:  pes.c
    untracked:  .DS_Store
    untracked:  Makefile
    untracked:  test_objects
    untracked:  tree.h
    untracked:  test_objects.c
    untracked:  commit.c
    untracked:  test_tree
    untracked:  README.md
    untracked:  index.c
    untracked:  test_sequence.sh
    untracked:  .gitignore
    untracked:  test_tree.c
    untracked:  compile_flags.txt
    untracked:  pes.h
    untracked:  commit.h
    untracked:  tree.c
```

Screenshot 3B — capture the index file contents

```
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % cat .pes/index

  100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776766066 6 file1.txt
  100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776766066 6 file2.txt
```

PHASE 4:

```
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % echo "Hello" > hello.txt
  ./pes add hello.txt
  ./pes commit -m "Initial commit"
  echo "World" >> hello.txt
  ./pes add hello.txt
  ./pes commit -m "Add world"
  echo "Goodbye" > bye.txt
  ./pes add bye.txt
  ./pes commit -m "Add farewell"
  ./pes log

  Committed: c65fc1be0410... Initial commit
  Committed: 7cee187aa8b0... Add world
  Committed: cd99e8c6fd2d... Add farewell
  commit cd99e8c6fd2d50d8e2c040a88c99dd80060fa9a4c37ab0c13f7968265b4350aa
  Author: PES User <pes@localhost>
  Date:   1776766205

      Add farewell

  commit 7cee187aa8b00e5f2599b971f696b1b450f811cb4c677046b1573e02a470e34b
  Author: PES User <pes@localhost>
  Date:   1776766205

      Add world

  commit c65fc1be0410b41a0827239123462cf9e00d41737579294a6871202c004f4bf1
  Author: PES User <pes@localhost>
  Date:   1776766205

      Initial commit
```

Screenshot 4A — capture ./pes log showing all 3 commits
Screenshot 4B — capture the full object store listing

```
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % find .pes -type f | sort

  .pes/HEAD
  .pes/index
  .pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
  .pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
  .pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
  .pes/objects/7c/ee187aa8b00e5f2599b971f696b1b450f811cb4c677046b1573e02a470e34b
  .pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
  .pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
  .pes/objects/c6/5fc1be0410b41a0827239123462cf9e00d41737579294a6871202c004f4bf1
  .pes/objects/cd/99e8c6fd2d50d8e2c040a88c99dd80060fa9a4c37ab0c13f7968265b4350aa
  .pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
  .pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
  .pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
  .pes/refs/heads/main
```

Screenshot 4C — capture both outputs

```
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % cat .pes/refs/heads/main
  cat .pes/HEAD

  cd99e8c6fd2d50d8e2c040a88c99dd80060fa9a4c37ab0c13f7968265b4350aa
  ref: refs/heads/main
```

Final Screenshot:

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % make test-integration

=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 606fa0560aed... Initial commit

Log after first commit:
commit 606fa0560aed930ceb586066c8821055f04e99fee9d1cc3b7dd8b5a7dc30511f
Author: PES User <pes@localhost>
Date:   1776766406

    Initial commit


--- Second Commit ---
Committed: 783353d17045... Update file.txt

--- Third Commit ---
Committed: 09d4e2a53c5e... Add farewell

--- Full History ---
commit 09d4e2a53c5e0cf85f0272d333934b3e61bfdb4cd4c66ab498cb0aecbeb11f13
Author: PES User <pes@localhost>
Date:   1776766406

    Add farewell

commit 783353d1704511baf9af82fd9d6d766425c71a92dd80c9360a44329981da7014
Author: PES User <pes@localhost>
Date:   1776766406

    Update file.txt

commit 606fa0560aed930ceb586066c8821055f04e99fee9d1cc3b7dd8b5a7dc30511f
Author: PES User <pes@localhost>
Date:   1776766406

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
09d4e2a53c5e0cf85f0272d333934b3e61bfdb4cd4c66ab498cb0aecbeb11f13

--- Object Store ---
Objects created:
      10
.pes/objects/09/d4e2a53c5e0cf85f0272d333934b3e61bfdb4cd4c66ab498cb0aecbeb11f13
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/60/6fa0560aed930ceb586066c8821055f04e99fee9d1cc3b7dd8b5a7dc30511f
.pes/objects/78/3353d1704511baf9af82fd9d6d766425c71a92dd80c9360a44329981da7014
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs %
```